Perpendicular lines

1 Two lines that intersect each other at an angle of 90°.

2 -1

3

a No **b** No **c** Yes **d** Yes

e No **f** Yes **g** Yes **h** No

4

a $m = -\frac{36}{5}$ **b** $m = 10$

5

a $-\frac{1}{6}$ **b** $-\frac{1}{6}$ **c** -8 **d** 1

6

a $-\frac{8}{7}$ **b** $\frac{8}{7}$ **c** No

7

a $x = -8$ **b** $y = -8$ **c** $y = 2x + 6$ **d** $y = -\frac{x}{6} + 10$

e $y = -\frac{x}{6} - 4$

8

a $-\frac{1}{5}$ **b** $y = -\frac{x}{5} + 1$

9

a $\frac{-1}{10}$ **b** $(5, 10)$

c $x + 10y - 105 = 0$

Applications

10

a

i 1 **ii** -1

b Yes, because their gradients multiply to -1.

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a

i $m = -1$ **ii** $m = 1$ **iii** $m = -1$ **iv** $m = 1$

b Square

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